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### **Tutorial Sheet 6**

#### **Problem Statement 1**

A company is deploying a smart home automation system with three components and the given infrastructure elements:

- **DeviceManager (Tiny VM)** - Manages IoT devices with latency  $\leq 100\text{ms}$  and bandwidth  $\geq 1\text{ Mbps}$ .
- **DataProcessor (Large VM)** - Handles data storage and processing with latency  $\leq 150\text{ms}$  and bandwidth  $\geq 3\text{ Mbps}$ .
- **UserInterface (Small VM)** - Provides user interaction, requiring latency  $\leq 200\text{ms}$  and bandwidth  $\geq 2\text{ Mbps}$ .

#### **Infrastructure Elements:**

- FogNode1: ₹2000/month, 2 vCPUs, 4 GB RAM, 40 GB HDD.
- FogNode2: ₹4000/month, 4 vCPUs, 8 GB RAM, 80 GB HDD.
- Cloud: Unlimited resources, VMs priced (Tiny: ₹500/month, Small: ₹1000/month, Large: ₹3000/month).

Using the concepts in FogTorch calculate the following:

- a. Resource consumption (CPU, RAM, storage per node).
- b. Verify QoS assurance (latency and bandwidth compliance).
- c. Compute cost of deployment.
- d. Optimize deployment to meet all requirements within ₹8500/month.

#### **Problem Statement 2**

A city authority is developing a fog-based Smart Traffic Control System to enhance road safety, monitor vehicles, and optimize traffic flow. This system has three modules: SensorModule, which collects encrypted, real-time data from IoT devices like road sensors and cameras with latency under 100ms; ProcessingModule, which uses moderate resources (2 vCPUs, 4 GB RAM, 40 GB HDD) to filter and aggregate data while ensuring privacy; and DecisionModule, which requires higher computational power (4 vCPUs, 8 GB RAM, 80 GB HDD) for advanced analytics and machine learning to deliver real-time traffic management feedback and commands.

- a. Design an architecture based on the three-layer model (Device, Fog, and Cloud layers) to handle the data management issues of the system.

- b. Draw an interaction diagram illustrating how the modules will communicate

### Problem Statement 3

An organization is deploying IoT-enabled fog nodes to monitor various small-scale edge scenarios in remote locations. The edge scenarios include:

- a. Flood warning system is deployed where IoT sensors monitor water levels and rainfall to detect flood risks in real-time. Which ML model may be suitable for anomaly detection and for forecasting flood severity.
- b. Wildfire Detection is deployed where thermal sensors and drones monitor temperature and smoke levels in forests. Which ML model may be suitable for classification of trees and for image processing from drone footage.
- c. Smart Home Intrusion Detection is deployed where Motion sensors and cameras detect unauthorized movements in homes. Which ML model may be suitable for anomaly detection and for analyzing video streams.
- d. Healthcare Monitoring System is deployed where IoT-enabled body sensors monitor vital signs to detect health anomalies. Which ML model may be suitable for categorizing health patterns and for finding anomalies in medical data.
- e. Energy Theft Prevention in Smart Grids is deployed where IoT devices track and analyze electricity consumption to identify unusual activities. Which ML model may be suitable for trend prediction and categorizing anomalies.

Identify and justify suitable machine learning models for each scenario based on:

- Low memory requirements.
- Fast processing capabilities at edge fog nodes.

### Problem Statement 4

A factory seeks to implement a Fog Engine architecture to monitor equipment health in real time. The system comprises three modules:

- **DataCollector:** Gathers sensor data (e.g., vibration, temperature) with latency <50ms.
  - **RealTimeAnalyzer:** Processes the data using moderate resources (e.g., 2 vCPUs, 4 GB RAM) to detect anomalies.
  - **AlertGenerator:** Issues maintenance alerts quickly (<100ms) using higher computing capacity (e.g., 4 vCPUs, 8 GB RAM).
- a. Outline an FEA that integrates Fog Engine Architecture
  - b. Describe the data flow in the give scenario and using above architecture from sensor collection to real-time anomaly detection and alert dispatching.

## Problem Statement 5

A metropolitan environmental agency is deploying an Urban Air Quality Monitoring System (UAQMS) that gathers real-time pollutant data (e.g., PM<sub>2.5</sub>, CO<sub>2</sub>, ozone) using thousands of distributed IoT sensors. Fog Engine nodes are installed at neighborhood levels to promptly process sensor data for issuing local alerts (e.g., warning of hazardous air conditions) and to support rapid, on-site decision-making. At the same time, comprehensive trend analysis and long-term policy planning are performed centrally in the cloud. However, the system must determine:

- When local, low-latency analytics at Fog Engine nodes suffice,
  - When to offload heavy or aggregated data analytics to the cloud, and
  - How processing tasks can be shared among multiple Fog Engine nodes in high-density sensor areas.
- a. Propose an integrated system architecture combining Fog Engine and cloud services. The integrated system representation should cover all the stages of data analytics function both at fog level and cloud level.
  - b. Define the criteria (e.g., latency thresholds, data volume, computational resource availability, network bandwidth, energy constraints, urgency of anomaly detection) that dictate whether data should be processed locally at the Fog Engine, offloaded to the cloud for heavy analytics, or handled through peer-to-peer collaboration among Fog Engine nodes.
  - c. Describe a mechanism that enables multiple Fog Engine nodes to share processing loads in regions with high sensor density, ensuring both real-time response and fault tolerance.